Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

There are three men and seven women taking a dance class. Number of different ways in which each man is paired with a woman partner, and the four remaining women are paired into two pairs each of two is

(A) 105

(B) 315

(C) 630

(D) 450

Q.2

If all permutations of the letters of the word AGAIN are arranged as in dictionary, then fiftieth word is

(A) NAAGI

(B) NAGAI

(C) NAAIG

(D) NAIAG

Q.3

How many numbers with no more than three digits can be formed using only the digits 1 through 7 with no digit used more than once in a given number?

(A) 259

(B) 249

(C) 257

(D) 252

Q.4

In a 12 - storey house ten people enter a lift cabin. It is known that they will left in groups of 2,3 and 5 people at different storeys. The number of ways they can do so if the lift does not stop to the second storey is

(A) 78

(B) 112

(C) 720

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(D) 132

Q.5

All the words that can be formed using alphabets $\mathrm{A}, \mathrm{H}, \mathrm{L}, \mathrm{U}$ and R are written as in a dictionary $(\mathrm{no}$ alphabet is repeated). Rank of the word RAHUL is

(A) 71

(B) 72

(C) 73

(D) 74

Q.6

n' is selected from the set $\{1, 2, 3, \ldots, 100\}$ and the number $2^n + 3^n + 5^n$ is formed. Total number of ways of selecting ' n ' so that the formed number is divisible by 4 , is equal to

(A) 50

(B) 49

(C) 48

(D) None of these

Q.7

5 - digit numbers are to be formed using $2, 3, 5, 7,$
9 without repeating the digits. If p be the number of such numbers that exceed 20000 and q be the number of those that lie between 30000 and 90000 , then $p : q$ is :

(A) $6 : 5$

(B) $3 : 2$

(C) $4 : 3$

(D) $5 : 3$

Q.8

If 12 persons are seated in a row, the number of ways of selecting 3 persons from them, so that no two of them are seated next to each other is

(A) 85

(B) 100

(C) 120

(D) 240

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Q.9

The number of permutations of the letters of the word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are

(A) 166674

(B) 169194

(C) 166680

(D) 181434

Q.10

A shop keeper sells threee varieties of perfumes and he has a large number of bottles of the same size of each variety in this stock. There are 5 places in a row in his show case. The number of different ways of displaying the three varieties of perfumes in the show case is

(A) 6

(B) 50

(C) 150

(D) None of these

Q.11

Total number of words that can be formed using all letters of the word FAILURE which neither begin with F nor end with E is equal to

(A) 3720

(B) 5040

(C) 3600

(D) 3480

Q.12

The number of different words which can be formed from the letters of the word LUCKNOW when the vowels always occupy even places is

(A) 120

(B) 720

(C) 400

(D) None of these

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Q.13

The number of different words that can be formed from the letters of the word 'PENCIL so that no two vowels are together is

(A) 120

(B) 260

(C) 240

(D) 480

Q.14

A closet has 5 pairs of shoes. The number of ways in which 4 ghoes can be chosen from it
so that there will be no complete pair is

(A) 80

(B) 160

(C) 200

(D) none of these

Q.15

The number of four -digit numbers strictly grater than 4321 that can be formed using the digits $0, 1, 2, 3, 4, 5$ (repetition of digits is allowed) is:

(A) 288

(B) 306

(C) 360

(D) 310

Q.16

There are seven greeting cards; each of different colour, and seven envelopes of the same seven colours. The number of ways in which the cards can be put in the envelopes, so that exactly four of the cards go into the envelopes of the right colours, is

(A) 35

(B) 70

(C) 140

(D) none of these

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Q.17

The numbers of arrangements of the letters of the word SALOON, if the two O°s do not come together, is

(A) 360

(B) 720

(C) 240

(D) 120

Q.18

In a city no two persons have identical set of teeth and there is no person without a tooth. Also no person has more than 32 teeth. If we disregard the shape and size of tooth and consider only the positioning of the teeth, then the maximum population of the city is

(A) 2^{32}

(B) $(32)^2 - 1$

(C) $2^{32} - 1$

(D) 2^{32-1}

Q.19

The number of times the digit 3 will be written when listing the integers from 1 to 1000 is

(A) 269

(B) 300

(C) 271

(D) 302

Q.20

A dictionary is printed consisting of 7 lettered words only that can be made with the letter of the word CRICKET. If the words are printed in alphabetical order, as in an ordinary dictionary, then the number of words before the word CRICKET is

(A) 530

(B) 480

(C) 531

(D) 481

Answers & Solutions

Q.1 Answer:

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630

Solution:

$10 < \frac{3m}{7w} 3$ **women can be selected in** ${}^{7}C_{3}$ **ways and can be paired with 3 men in** $3!$ **ways. Remaining 4 women can be grouped into two couples in** $4!/(2!2! \cdot 2!) = 3$
Therefore, total $= {}^{7}C_{3}.3!.3 = 630$

Q.2 Answer:

NAAIG

Solution:

Starting with the letter A**, and arranging the other four letters, there are** $4! = 24$ **words. These are the first 24 words. Then starting with** G**, and arranging** $\mathrm{A}, \mathrm{A}, \mathrm{I}$**, and** N **in different ways, there are** $\frac{4!}{2!1!1!} = \frac{24}{2} = 12$ **words. Hence, total 36 words. Next, the 37 th word starts with** I**. There are 12 words starting with I. This accounts up to the 48 th word. The 49th word is NAAGI. The 50 th word is NAAIG.**

Q.3 Answer:

259

Solution:

(i) Number of 3 digit nos. $= 7 \times 6 \times 5 = 210$
(ii) Number of 2 digit nos. $= 7 \times 6 = 42$
(iii) Number of 1 digit no. $= 7$ **Total number of nos.** $= 210 + 42 + 7 = 259$

Q.4 Answer:

720

Solution:

Leaving the ground floor and second floor, their are 10 floors in which three groups of people can left the lift cabin in ${}^{10}\mathrm{P}_{3}$ **ways,**
i.e. 720 ways.

Q.5 Answer:

74

Solution:

No. of words starting with A **are** $4! = 24$ **No. of words starting with** H **are** $4! = 24$ **No. of words starting with** L **are** $4! = 24$ **These account for 72 words Next word is RAHLU and the** 74th **word RAHUL**

Q.6 Answer:

49

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Solution:

If n is odd, $3^n = 4\lambda_1 - 1, 5^n = 4\lambda_2 + 1$
$\Rightarrow 2^n + 3^n + 5^n$ is not divisible by 4, as $2^n + 3^n + 5'$
will be in the form of $4\lambda + 2$. Thus, total number of ways of selecting ' n ' is equal to 49.

Q.7 Answer:

$5 : 3$

Solution:

p :	T T H	T H	H	T	0	place
	5	4	3	2	1	ways

Total no. of ways $= 5! = 120$ Since all numbers are $> 20,000$
$\therefore$ all numbers $2, 3, 5, 7, 9$ can come at first place. TTH TH H T 0 place q: $3 \quad 4 \quad 321$ ways
Total no. of ways $= 3 \times 4! = 72$ ($\because$ 2 and 9 can not be put at first place) So, $p : q = 120 : 72 = 5 : 3$

Q.8 Answer:

120

Solution:

The number of ways of selecting 3 persons from 12 people under the given conditon :
- Number of ways of arranging 3 people among people seated in a row, so that no two of them are consecutive
= Number of ways of choosing 3 places out of the $10[8$ in between and 2 extremes $]$
$= {}^{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 5 \times 3 \times 8 = 120$

Q.9 Answer:

169194

Solution:

Total number of permutations $= \frac{9!}{2!}$ Number of those containing 'HIN' $= 7!$
Number of those containing 'DUS' $= \frac{7!}{2!}$ Number of those containing 'TAN' $= 7!$ Number of those containing 'HIN' and 'DUS' $= 5!$ Number of those containing 'HIN' and 'TAN' $= 5!$ Number of those containing 'TAN' and 'DUS' $= 5!$ Number of those containing 'HIN', 'DUS' and 'TAN' $= 3!$
Required number $= \frac{9!}{2!} - \left(7! + 7! + \frac{7!}{2}\right) + 3 \times 5! - 3! = 169194.$

Q.10 Answer:

150

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Solution:

Number of ways of displaying without restriction $= 3^5$
Number of ways in which all places are occupied by single variety $= 3 \times 1$ **Number of ways in which all places are occupied by two different varieties** $= {}^3\mathrm{C}_2 \left[2^5 - 2\right]$ $[\because$ **There are two ways when all places will be occupied by single variety each.]** $\therefore$ **No. of way of displaying all three varieties** $= 3^5 - 3 - {}^3\mathrm{C}_2 \left(2^5 - 2\right) = 150$

Q.11 Answer:

3720

Solution:

Number of words beginning with $'F' = 6!$
Number of words ending with $'E' = 6!$
Number of words beginning with $'F'$**, and ending with** $'E' = 5!$
Total number of words that neither begins with 'E' nore end with 'F'

$$7! - (6! + 6! - 5!)$$
$$= 5040 - 1440 + 120$$
$$= 3720$$

Q.12 Answer:

720

Solution:

In LUCKNOW, number of letters $= 7$ **(all distinct), vowels** $(\mathrm{U}, \mathrm{O}) = 2$**, consonants** $(\mathrm{L}, \mathrm{C},, \mathrm{K}, \mathrm{N}, \mathrm{W}) = 5$
There are 7 letters so 7 places are required. Number of even places are 3 and vowels are 2, so 2 vowels can be placed in ${}^3C_2 \cdot 2! = 6$ **and 5 consonants can be placed in remaining places in 5! ways. Hence, number of words** $= 6.5! = 720.$

Q.13 Answer:

480

Solution:

1. There is no restriction on the arrangement of the letters.
The six different letters can be arranged in $6_{p_6} = \underline{\underline{6!}} = 720$ **ways.**
Hence the total number of words formed $= 720.$
2. Vowels occur together:
The vowels are E **and** I**. Regarding them as one letter, the 5 letters can be arranged in** $\underline{\underline{5!}} = 120$ **ways.**
These two vowels can be arranged amongst themselves in $\underline{\underline{2!}} = 2$ **ways.**
$\therefore$ **The total number of words** $= 2 \times 120 = 240.$

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But according to our question n o two vowels are together

So answer $= 720 - 240 = 480$

Q.14 Answer:

80

Solution:

Select 4 pairs in ${}^{3}C_{4} = 5$ **ways**

Now select exactly one shoe from each of the pair selected in $\left({}^{2}C_{1}\right)^{4}$ **ways**

$\Rightarrow$ **Required answers** $= 5 \times 16 = 80$

Q.15 Answer:

310

Solution:

(1) The number of four-digit numbers Strating with 5 is equal to $6^{3} = 216$

(2) Starting with 44 and 55 is equal to $36 \times 2 = 72$

(3) Starting with 433,434 and 435 is equal to $6 \times 3 = 18$

(4) Remaining numbers are $4322, 4323, 4324, 4325$ **is equal to 4 so total number are**

$716 + 72 + 18 + 4 = 310$

Q.16 Answer:

70

Solution:

Select four cards in ${}^{7}C_{4}$ **ways to put them in correct envelopes then remaining all will go wrong in 2 ways (since number of no matches for three objects is 2).**

$\Rightarrow$ **Required answer** $= 2 \times {}^{7}C_{4} = 70$

Q.17 Answer:

240

Solution:

Total number of arrangements are $\frac{6!}{2!} = 360$

The number of ways in which O' **s come together** $= 5! = 120$

Hence required number of ways $= 360 - 120 = 240$

Q.18 Answer:

$2^{32} - 1$

Solution:

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We have, 32 places for teeth.

For each place we have two choices either there is a tooth or there is no tooth.

Therefore the number of ways to fill up these places is 2^{32}

As there is no person without a tooth, the maximum population is $2^{32}-1$

Q.19 Answer:

300

Solution:

Since 3 does not occur in 1000 , we have

Count the number of times 3 occurs when we list the integers from 1 to 999

Any number between 1 and 999 is of the form xyz, where $0 \leq x, y, z \leq 9$

First count numbers in which 3 occurs exactly once.

Since 3 can occur at one place in ${}^{3}C_{1}$ ways

$\therefore {}^{3}C_{1}(9 \times 9) = 3 \times 9^{2}$

Next, 3 can occur in exactly two places in $\left({}^{3}C_{2}\right)(9) = 3 \times 9$

Lastly, 3 can occur in all three digits in one number only.

Hence, the number of times 3 occurs is

$$1 \times \left(3 \times 9^{2}\right) + 2 \times (3 \times 9) + 3 \times 1 = 300$$

Q.20 Answer:

530

Solution:

In the alphabetic order the letters are $\mathrm{C}, \mathrm{C}, \mathrm{E}, \mathrm{I}, \mathrm{K}, \mathrm{R}, \mathrm{T}$. Words beginning with C will be printed at first. Our word also begins with C.

The no. of words beginning with C and having C or E and I or R in second place $= 4 \times 5!$

The no. of words begin with CR and having C or E in the third place $= 2 \times 4!$ The no. of words begin with CRIC and having E in the fifth place $= 2!$ After printing all three words, the word CRICKET will be printed.

$\therefore$ The required place for the word CRICKET is $= (4 \times 5! + 2 \times 4! + 2!) + 1 = 530 + 1 = 531^{\text{th}}$ place

$\therefore$ The word before it are 530.

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